

CiteBase Engineering Whitepaper

1. Executive Summary

CiteBase is a high-throughput, multi-tenant document intelligence engine designed to process complex documents with zero hallucinations. By strictly grounding generated responses in verified document passages, the system eliminates extrapolation and provides verbatim citations with page references for enterprise compliance.

1.1 System Objectives

Primary operational objectives include achieving sub-500ms end-to-end retrieval latency, maintaining over 95% precision on page citations, and supporting concurrent document ingestion across distributed worker threads without blocking API responsiveness.

2. Technical Architecture

2.1 Smart Layout-Aware Chunking

Traditional RAG systems slice text indiscriminately using character windows, frequently bisecting critical sentences and separating table headers from their underlying data values. CiteBase employs layout-aware boundary detection that parses font hierarchies, markdown headers, and semantic paragraph blocks before applying recursive character splitting. Contextual breadcrumbs are prepended to each chunk to preserve document hierarchy.

2.2 Hybrid Retrieval and Reranking

Retrieval employs a dual-channel architecture fusing dense semantic embeddings from all-MiniLM-L6-v2 with BM25Okapi sparse keyword retrieval via Reciprocal Rank Fusion (RRF). Initial candidate pools (top 20) are then passed through a cross-encoder model (ms-marco-MiniLM-L-6-v2) for deep relevance scoring before the top 5 passages are provided to Google Gemini.